Triple integrals and vector derivatives

1. Adding contributions throughout a solid

For a double integral, a small piece of the region has area dA. For a triple integral, a small rectangular piece of the solid has volume

$$\Delta V = \Delta x\, \Delta y\, \Delta z.$$

If F is approximately constant on that small piece, its contribution is $F\,\Delta V$. Adding these contributions and taking the limit gives the **triple integral**

$$\iiint_E F(x,y,z)\,\mathrm{d}V.$$

Here E denotes the solid and dV denotes its small volume element. A sufficient condition used here is that F is continuous on a closed, bounded solid with a suitably regular boundary. All the solids and functions below satisfy this condition.

What should the integrand be?

$$\boxed{V = \iiint_E 1\,\mathrm{d}V} \qquad \text{and} \qquad \boxed{M = \iiint_E \rho(x,y,z)\,\mathrm{d}V}.$$

Integrating 1 adds volumes. Integrating a density ρ, measured in $\mathrm{kg/m^3}$, adds masses because $\mathrm{kg/m^3} \times \mathrm{m^3} = \mathrm{kg}$.

In Cartesian coordinates, we evaluate a triple integral by three ordinary integrations. In $\int\int\int F\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$, integrate first with respect to z, then y, then x. While integrating with respect to one variable, hold the other two fixed.

Problem 1. A rectangular material block

A block occupies

$$0 \le x \le 2, \qquad 0 \le y \le 1, \qquad 0 \le z \le 3,$$

where the coordinates are measured in metres.
(a) Find its volume using a triple integral.
(b) Its density is $\rho(x,y,z) = 2 + x$ $\mathrm{kg/m^3}$, where x is the numerical coordinate in metres. Find its mass.

(a) Volume: one integral for each coordinate

All the bounds are constant because the solid is a rectangular box:

$$\begin{aligned} V &= \int_0^2 \int_0^1 \int_0^3 1\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^2 \int_0^1 [z]_0^3\,\mathrm{d}y\,\mathrm{d}x = \int_0^2 \int_0^1 3\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^2 [3y]_0^1\,\mathrm{d}x = \int_0^2 3\,\mathrm{d}x = [3x]_0^2 = \boxed{6\ \mathrm{m^3}}. \end{aligned}$$

This agrees with length $\times$ breadth $\times$ height: $2 \times 1 \times 3 = 6$.

(b) Mass: include the density

The density is positive throughout the block: $2 \le 2 + x \le 4$. Use the same bounds but change the integrand:

$$\begin{aligned} M &= \int_0^2 \int_0^1 \int_0^3 (2 + x)\, dz\, dy\, dx \\ &= \int_0^2 \int_0^1 [(2 + x)z]_0^3\, dy\, dx \\ &= \int_0^2 \int_0^1 3(2 + x)\, dy\, dx \\ &= \int_0^2 [3(2 + x)y]_0^1\, dx = 3\int_0^2 (2 + x)\, dx \\ &= 3\left[2x + \frac{x^2}{2}\right]_0^2 = 3(4 + 2) = \boxed{18 \text{ kg}}. \end{aligned}$$

As a check, the average density is $18/6 = 3$ kg/m^3, which lies between the smallest and largest densities.

2. Finding bounds from the shape

Problem 2. A tetrahedron below a plane

Find the volume of the solid

$$E = \{(x, y, z) : x \ge 0,\ y \ge 0,\ z \ge 0,\ x + 2y + z \le 4\}.$$

1. Identify the floor and roof

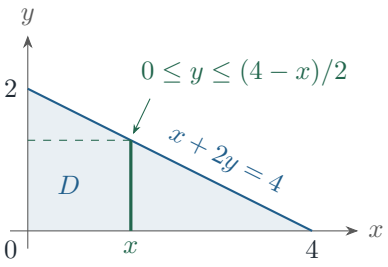
The coordinate planes restrict the solid to the first octant, where all three coordinates are nonnegative. The plane $x + 2y + z = 4$ cuts the axes at

$$(4, 0, 0), \qquad (0, 2, 0), \qquad (0, 0, 4).$$

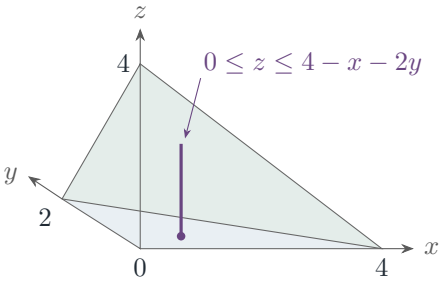
For fixed x, y, solve the plane equation for z:

$$\boxed{\underbrace{0}_{\text{floor}} \le z \le \underbrace{4 - x - 2y}_{\text{roof}}}.$$

The roof must lie at or above the floor. Therefore $4 - x - 2y \ge 0$, or $x + 2y \le 4$. This determines the projection onto the xy-plane.



Projection onto the xy-plane



A vertical segment in the solid

The green strip in the projection fixes an x and lets y vary. Above each point of that strip, a vertical segment runs from $z = 0$ to the sloping roof.

2. Read the remaining two bounds

In the projection,

$$D = \{(x, y) : x \geq 0,\ y \geq 0,\ x + 2y \leq 4\}.$$

The entire triangle extends from $x = 0$ to $x = 4$. Once x is fixed, y starts at the x-axis and ends on the line $x + 2y = 4$. Hence

$$\boxed{0 \leq x \leq 4, \qquad 0 \leq y \leq \frac{4 - x}{2}}.$$

Putting the bounds together in the order $dz\,dy\,dx$,

$$\boxed{V = \int_0^4 \int_0^{(4-x)/2} \int_0^{4-x-2y} 1 \,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x}.$$

3. Evaluate from the inside out

First integrate with respect to z:

$$\int_0^{4-x-2y} 1 \,\mathrm{d}z = [z]_0^{4-x-2y} = 4 - x - 2y.$$

Now integrate with respect to y, treating x as fixed:

$$\begin{aligned}\int_0^{(4-x)/2} (4 - x - 2y) \,\mathrm{d}y &= \Big[(4 - x)y - y^2\Big]_0^{(4-x)/2} \\ &= \frac{(4 - x)^2}{2} - \frac{(4 - x)^2}{4} = \frac{(4 - x)^2}{4}.\end{aligned}$$

Finally,

$$\begin{aligned}V &= \frac{1}{4} \int_0^4 (4 - x)^2 \,\mathrm{d}x = \frac{1}{4} \left[-\frac{(4 - x)^3}{3} \right]_0^4 \\ &= \frac{1}{4} \left(0 + \frac{64}{3} \right) = \boxed{\frac{16}{3} \text{ cubic units}}.\end{aligned}$$

Connection with a double integral

The first integration adds a vertical column of height $4 - x - 2y$. Thus

$$\iiint_E 1 \,\mathrm{d}V = \iint_D (4 - x - 2y) \,\mathrm{d}A.$$

Both expressions calculate the same volume. The triple integral integrates through the height explicitly; the double integral uses that height as its integrand.

3. A moving point and its derivatives

A **vector function** $\mathbf{r}(t) = (x(t), y(t), z(t))$ gives three component functions of one variable. Its derivative is defined by

$$\mathbf{r}'(t) = \lim_{h \to 0} \frac{\mathbf{r}(t + h) - \mathbf{r}(t)}{h},$$

when this limit exists. The vector limit is taken component by component. If $\mathbf{r}$ represents position at time t, this derivative gives velocity:

Position, velocity, acceleration and speed

$$\mathbf{v}(t) = \mathbf{r}'(t) = (x'(t), y'(t), z'(t)),$$
$$\mathbf{a}(t) = \mathbf{r}''(t) = (x''(t), y''(t), z''(t)),$$
$$\text{speed} = \|\mathbf{v}(t)\| = \sqrt{(x')^2 + (y')^2 + (z')^2}.$$

When $\mathbf{r}'(t) \neq \mathbf{0}$, the unit tangent in the direction of increasing t is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}.$$

Velocity gives both direction and rate of motion. Speed is a nonnegative number. The fixed Cartesian unit vectors do not change when we differentiate.

Problem 3. Position, motion and tangent direction

A particle has position

$$\mathbf{r}(t) = (t, t^2, 2t), \qquad 0 \leq t \leq 2,$$

where t is the numerical time in seconds and the components give position in metres. Find its position, velocity, acceleration, speed and unit tangent at $t = 1$.

1. Differentiate before substituting the time

$$\mathbf{r}(1) = (1, 1, 2) \text{ m},$$
$$\mathbf{v}(t) = \mathbf{r}'(t) = (1, 2t, 2) \implies \mathbf{v}(1) = (1, 2, 2) \text{ m/s},$$
$$\mathbf{a}(t) = \mathbf{r}''(t) = (0, 2, 0) \implies \mathbf{a}(1) = (0, 2, 0) \text{ m/s}^2.$$

Substituting $t = 1$ first would turn the position into a constant vector and lose its time dependence. Therefore we differentiate the function first.

2. Take the magnitude for speed

$$\text{speed at } t = 1 = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = \boxed{3 \text{ m/s}}.$$

3. Divide by the magnitude for unit direction

$$\boxed{\mathbf{T}(1) = \frac{(1, 2, 2)}{3} = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)}.$$

Its magnitude is $\sqrt{1/9 + 4/9 + 4/9} = 1$, as required.

4. Change of temperature in a chosen direction

A **scalar field** assigns one number to each point, such as a temperature $T(x, y, z)$. Its partial derivatives measure change along the coordinate directions. Collecting them produces the **gradient vector**:

$$\boxed{\nabla T = (T_x, T_y, T_z)}.$$

This differs from $\mathbf{r}'(t)$: a position function is differentiated with respect to one parameter t; a scalar field has partial derivatives with respect to its spatial coordinates.

Problem 4. Temperature, direction and a tangent plane

The temperature near a point is modelled by

$$T(x, y, z) = 20 + x^2 + 2y - z.$$

The coordinates are numerical values in metres and T is in degrees Celsius. Let $P = (1, 0, 1)$ and $Q = (3, 1, 3)$.
(a) Find the gradient at P.
(b) Find the temperature's rate of change at P, per metre travelled directly towards Q.
(c) Find the tangent plane at P to the surface on which the temperature is 20°C.

(a) Calculate the three partial derivatives

Holding the other variables fixed each time,

$$T_x = 2x, \qquad T_y = 2, \qquad T_z = -1.$$

Therefore

$$\nabla T(x, y, z) = (2x, 2, -1), \qquad \boxed{\nabla T(P) = (2, 2, -1)}.$$

The components have units of degrees Celsius per metre. For example, increasing z while holding x, y fixed decreases the temperature at 1°C per metre.

(b) First convert the direction into a unit vector

The vector pointing from P towards Q is

$$\mathbf{a} = Q - P = (3 - 1,\ 1 - 0,\ 3 - 1) = (2, 1, 2).$$

Its length is $\|\mathbf{a}\| = \sqrt{4 + 1 + 4} = 3$. The required unit vector is

$$\boxed{\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|} = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)}.$$

Normalisation matters: the question asks for change *per unit distance*, not per step of length 3.

Directional derivative

For a differentiable scalar field and a unit vector $\mathbf{u} = (u_1, u_2, u_3)$,

$$\boxed{D_{\mathbf{u}}T(P) = \nabla T(P) \cdot \mathbf{u}}.$$

To see why, move along $\mathbf{r}(s) = P + s\mathbf{u}$. Because $\|\mathbf{u}\| = 1$, s is signed distance along the line (the distance from P is $|s|$). The chain rule gives

$$\frac{d}{ds} T(P + s\mathbf{u})\bigg|_{s=0} = T_x(P)u_1 + T_y(P)u_2 + T_z(P)u_3.$$

Apply the formula:

$$\begin{aligned} D_{\mathbf{u}}T(P) &= (2, 2, -1) \cdot \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right) \\ &= \frac{4}{3} + \frac{2}{3} - \frac{2}{3} = \boxed{\frac{4}{3}\ ^\circ\text{C/m}}. \end{aligned}$$

The positive sign means that temperature initially increases when we move from P towards Q.

(c) Use the gradient as a normal to a level surface

First check that P lies on the stated surface:

$$T(1,0,1) = 20 + 1 + 0 - 1 = 20.$$

A surface $T =$ constant is called a **level surface**. Along any curve on that surface, T stays constant, so its rate of change in a tangent direction is zero. The chain rule then says that the gradient is perpendicular to every tangent direction.

Tangent plane to a regular level surface

If T has continuous first partial derivatives near $P = (x_0, y_0, z_0)$ and $\nabla T(P) \neq \mathbf{0}$, the tangent plane to $T = T(P)$ is

$$\boxed{T_x(P)(x - x_0) + T_y(P)(y - y_0) + T_z(P)(z - z_0) = 0}.$$

The vector $\nabla T(P)$ is a normal vector to this plane.

The polynomial T has continuous first partial derivatives, and $\nabla T(P) = (2, 2, -1) \neq \mathbf{0}$. Hence

$$\begin{aligned} 2(x-1) + 2(y-0) - (z-1) &= 0, \\ 2x - 2 + 2y - z + 1 &= 0, \\ \boxed{2x + 2y - z = 1}&. \end{aligned}$$

Check that P lies on the plane: $2(1) + 2(0) - 1 = 1$. The plane approximates the curved temperature surface near P; it need not coincide with that surface away from P.

Further calculations with the same temperature field

A direct check of the directional derivative

Substitute the unit-speed line

$$(x, y, z) = \left(1 + \frac{2s}{3}, \frac{s}{3}, 1 + \frac{2s}{3}\right)$$

into T:

$$\begin{aligned} T(P + s\mathbf{u}) &= 20 + \left(1 + \frac{2s}{3}\right)^2 + \frac{2s}{3} - \left(1 + \frac{2s}{3}\right) \\ &= 20 + \frac{4s}{3} + \frac{4s^2}{9}. \end{aligned}$$

Therefore its derivative at $s = 0$ is $4/3$, agreeing with the gradient calculation. This is the chain rule written out for this particular line.

Direction of greatest increase

For a unit direction $\mathbf{u}$, the dot-product formula gives

$$D_{\mathbf{u}}T(P) = \|\nabla T(P)\| \cos\theta,$$

where θ is the angle between $\mathbf{u}$ and $\nabla T(P)$. The largest value occurs at $\theta = 0$, when the two directions agree. Here

$$\|\nabla T(P)\| = \sqrt{2^2 + 2^2 + (-1)^2} = 3.$$

Thus the greatest rate of increase is

$$\boxed{3\ ^\circ\text{C/m}}, \qquad \text{in direction } \boxed{\left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)}.$$

The direction towards Q has rate $4/3$, so it is not the direction of greatest increase.